

Multiple Input Multiple Output Wireless Communications: Introduction to MIMO Communications, MIMO system Model, MIMO Zero Forcing Receiver, MIMO MMSE Receiver, Singular Value decomposition of MIMO Channel, SVD and MIMO capacity, Alamouti and Space-Time Codes, Nonlinear MIMO receiver: V-Blast, MIMO Beamforming.

[Text1:6.1,6.2, 6.3, 6.4, 6.5, 6.6, 6.8, 6.9, 6.10]

1. With a neat diagram explain MIMO system model. (10 marks)
2. Explain MIMO zero forcing Receiver. (10 marks)
3. MIMO MMSE receiver. (10 marks)
4. Explain singular value decomposition of MIMO channel. (10 marks)
5. Note on a) SVD and MIMO capacity
b) Nonlinear MIMO receiver: V-Blast, MIMO Beamforming.

1. With a neat diagram explain MIMO system model. (10 marks)

Consider a wireless MIMO a wireless system with multiple antennas at both the transmitter and the receiver.

- t: Number of transmit antennas.
- r: Number of receive antennas.
- This is called an $r \times t$ MIMO system.

The Transmit Side

$x_1, x_2, \dots, x_t$: These are the individual symbols (pieces of data) being sent.

Each symbol x_i is transmitted from its own dedicated antenna (the i -th transmit antenna).

These symbols are grouped together into a Transmit Vector, $\mathbf{x}$:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}$$

This vector $\mathbf{x}$ is a compact mathematical representation of what is being sent from all antennas at a given moment.

The Receiver Side

$y_1, y_2, \dots, y_r$: These are the individual symbols received at each antenna.

Each receive antenna hears a combination of all the transmitted signals, altered by the wireless channel.

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}$$

This vector $\mathbf{y}$ represents the signals observed at all receive antennas.

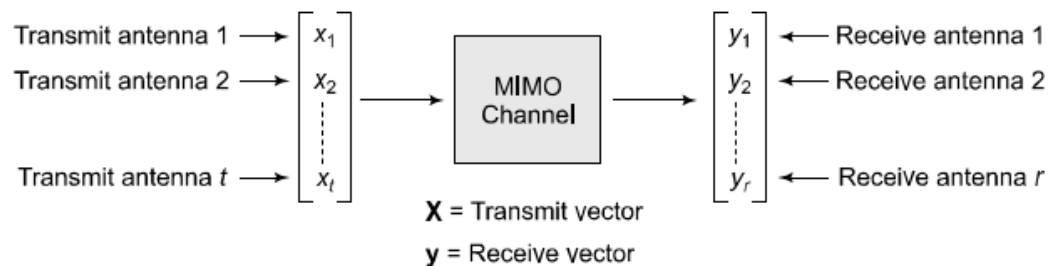


Figure 6.2 MIMO system input-output schematic

Let the complex channel coefficient h_{ij} between i^{th} transmit channel to j^{th} receive channel.

$$\mathbf{H} = \begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}$$

The net MIMO channel can be represented as

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}}_{\mathbf{n}}$$

$$\mathbf{y} = \mathbf{H} \mathbf{x} + \mathbf{n}$$

The signal at the first receive antenna y_1 is:

$$y_1 = h_{11}x_1 + h_{12}x_2 + \dots + h_{1t}x_t + n_1$$

The signal at the first receive antenna y_2 is:

$$y_2 = h_{21}x_1 + h_{22}x_2 + \dots + h_{2t}x_t + n_2$$

Case1: SISO (single input and single output antenna).

$$y = hx + n$$

Case2: SIMO (Single input and multiple output)

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_r \end{bmatrix}}_{\mathbf{h}} x + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}}_{\mathbf{n}}$$

Case3: MISO (Multiple input and single output)

$$y = \underbrace{\begin{bmatrix} h_1 & h_1 & \dots & h_t \end{bmatrix}}_{\mathbf{h}^T} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}}_{\mathbf{x}} + n$$

Case4: MIMO (Multiple input and Multiple output)

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_r \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_r \end{bmatrix}}_{\mathbf{n}}$$

The covariance matrix of the noise $\mathbf{R}_n$ of the noise vector $\mathbf{n}$ is defined as

$$\mathbf{R}_n = E \{ \mathbf{n} \mathbf{n}^H \}$$

$$= E \left\{ \begin{bmatrix} n_1 \\ n_2 \\ \vdots \\ n_L \end{bmatrix} \begin{bmatrix} n_1^* & n_2^* & \dots & n_L^* \end{bmatrix} \right\}$$

$$\begin{aligned}
 &= \begin{bmatrix} E\{|n_1|^2\} & E\{n_1 n_2^*\} & \dots & E\{n_1 n_r^*\} \\ E\{n_2 n_1^*\} & E\{|n_2|^2\} & \dots & E\{n_2 n_r^*\} \\ \vdots & \vdots & \ddots & \vdots \\ E\{n_r n_1^*\} & E\{n_r n_2^*\} & \dots & E\{|n_r|^2\} \end{bmatrix} \\
 &= \begin{bmatrix} \sigma_n^2 & 0 & 0 & \dots & 0 \\ 0 & \sigma_n^2 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \sigma_n^2 \end{bmatrix} \\
 &= \sigma_n^2 \mathbf{I}_r
 \end{aligned}$$

He noises at different antennas is independent so

$$E\{n_i n_j^*\} = 0$$

The model is now extended to operate over discrete time instants, **indexed by k**:

$$\underbrace{\begin{bmatrix} y_1(k) \\ y_2(k) \\ \vdots \\ y_r(k) \end{bmatrix}}_{\mathbf{y}(k)} = \underbrace{\begin{bmatrix} h_{11} & h_{12} & \dots & h_{1t} \\ h_{21} & h_{22} & \dots & h_{2t} \\ \vdots & \vdots & \ddots & \vdots \\ h_{r1} & h_{r2} & \dots & h_{rt} \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1(k) \\ x_2(k) \\ \vdots \\ x_t(k) \end{bmatrix}}_{\mathbf{x}(k)} + \underbrace{\begin{bmatrix} n_1(k) \\ n_2(k) \\ \vdots \\ n_r(k) \end{bmatrix}}_{\mathbf{n}(k)}$$

The noise covariance matrix is given as

$$E\{\mathbf{n}(k) \mathbf{n}(l)^H\} = \sigma^2 \delta(k-l) \mathbf{I}_r$$

where the delta function $\delta(k-l) = 1$ if $k=l$ and 0 otherwise.

This noise process, which is uncorrelated across different antennas and time instants is termed as spatio temporally uncorrelated noise.

2. Explain MIMO zero forcing Receiver. (10 marks)

- The Zero-Forcing (ZF) receiver is a linear detection method used in MIMO systems to recover the transmitted signal vector $\mathbf{x}$ from the received signal vector $\mathbf{y}$.
- Its core principle is to completely eliminate the inter-symbol interference (ISI) caused by the channel mixing the data streams. It achieves this by applying the pseudo-inverse of the channel matrix $\mathbf{H}$ to the received signal.
- The derivation is based on minimizing the Least-Squares error between the received signal and the signal predicted by the channel model (MMSE).

The fundamental MIMO system equation is:

$$\mathbf{y} = \mathbf{H}\mathbf{x}$$

where:

- $\mathbf{y}$ is the $r \times 1$ received signal vector.
- $\mathbf{H}$ is the $r \times t$ channel matrix.
- $\mathbf{x}$ is the $t \times 1$ transmitted signal vector.

The goal of the receiver is to solve for the unknown transmitted vector $\mathbf{x}$ given $\mathbf{y}$ and $\mathbf{H}$.

- **Square Channel** ($r = t$): If $\mathbf{H}$ is square and invertible, the solution is straightforward: $\hat{\mathbf{x}} = \mathbf{H}^{-1}\mathbf{y}$

Least-Squares Estimation

For the over-determined case ($r > t$), the ZF receiver finds the estimate $\hat{\mathbf{x}}$ that minimizes the **squared error** between the actual received signal and the signal that would be received if $\hat{\mathbf{x}}$ were transmitted.

The error function (or cost function) is defined as:

$$f(\mathbf{x}) = \|\mathbf{y} - \mathbf{H}\mathbf{x}\|^2$$

This is known as the **Least-Squares error function**. The vector $\hat{\mathbf{x}}$ that minimizes this function is the **Least-Squares estimator**.

Step 1: Expand the Error Function

The error function is expanded as follows:

$$\begin{aligned} f(\mathbf{x}) &= (\mathbf{y} - \mathbf{H}\mathbf{x})^T(\mathbf{y} - \mathbf{H}\mathbf{x}) \\ &= \mathbf{y}^T\mathbf{y} - \mathbf{x}^T\mathbf{H}^T\mathbf{y} - \mathbf{y}^T\mathbf{H}\mathbf{x} + \mathbf{x}^T\mathbf{H}^T\mathbf{H}\mathbf{x} \end{aligned}$$

Since $\mathbf{y}^T\mathbf{H}\mathbf{x}$ is a scalar, it is equal to its own transpose: $\mathbf{y}^T\mathbf{H}\mathbf{x} = (\mathbf{y}^T\mathbf{H}\mathbf{x})^T = \mathbf{x}^T\mathbf{H}^T\mathbf{y}$. This allows us to simplify the expression to:

$$f(\mathbf{x}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{x}^T\mathbf{H}^T\mathbf{y} + \mathbf{x}^T\mathbf{H}^T\mathbf{H}\mathbf{x}$$

Step 2: Apply Vector Calculus to Find the Minimum

To find the minimum, we take the derivative of $f(\mathbf{x})$ with respect to the vector $\mathbf{x}$ and set it to zero. The text outlines key rules of vector derivatives:

- $\frac{\partial(\mathbf{c}^T \mathbf{x})}{\partial \mathbf{x}} = \mathbf{c}$ where $\mathbf{c}$ is a constant vector.
- $\frac{\partial(\mathbf{x}^T \mathbf{A} \mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}$ if $\mathbf{A}$ is a symmetric matrix ($\mathbf{A}^T = \mathbf{A}$). Note that $\mathbf{H}^T \mathbf{H}$ is always symmetric.

Applying these rules:

$$\begin{aligned} \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} &= \frac{\partial}{\partial \mathbf{x}}(\mathbf{y}^T \mathbf{y}) - \frac{\partial}{\partial \mathbf{x}}(2\mathbf{x}^T \mathbf{H}^T \mathbf{y}) + \frac{\partial}{\partial \mathbf{x}}(\mathbf{x}^T \mathbf{H}^T \mathbf{H} \mathbf{x}) \\ &= 0 - 2\mathbf{H}^T \mathbf{y} + 2\mathbf{H}^T \mathbf{H} \mathbf{x} \end{aligned}$$

Step 3: Solve for the Optimal Estimate $\hat{\mathbf{x}}$

Set the derivative to zero to find the minimizing value:

$$\begin{aligned} -2\mathbf{H}^T \mathbf{y} + 2\mathbf{H}^T \mathbf{H} \hat{\mathbf{x}} &= 0 \\ \mathbf{H}^T \mathbf{H} \hat{\mathbf{x}} &= \mathbf{H}^T \mathbf{y} \end{aligned}$$

Solving for $\hat{\mathbf{x}}$:

$$\hat{\mathbf{x}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{y}$$

Step 4: Generalize to Complex-Valued Signals

For complex matrices and vectors (indicated by the Hermitian transpose H instead of the regular transpose T), the final ZF solution becomes:

$$\hat{\mathbf{x}}_{ZF} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y}$$

Step 5: The Final Zero-Forcing Receiver

The Zero-Forcing receiver is formally defined by the equation:

$$\hat{\mathbf{x}}_{\text{ZF}} = \underbrace{(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H}_{\mathbf{F}_{\text{ZF}}} \mathbf{y}$$

- **ZF Receiver Matrix ($\mathbf{F}_{\text{ZF}}$):** The matrix $\mathbf{F}_{\text{ZF}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$ is known as the **pseudo-inverse** of the channel matrix $\mathbf{H}$. This is the matrix that effectively inverts the channel's effect.
- **How it works:** When applied to the received signal $\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$ (where $\mathbf{n}$ is noise), we get:

$$\hat{\mathbf{x}}_{\text{ZF}} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H (\mathbf{H}\mathbf{x} + \mathbf{n}) = \mathbf{x} + \underbrace{(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{n}}_{\text{Amplified Noise}}$$

The term $\mathbf{H}^H \mathbf{H}$ cancels out, perfectly removing the inter-stream interference. However, the noise term $\mathbf{n}$ is multiplied by the pseudo-inverse.

EXAMPLE 6.1

Compute the MIMO zero-forcing receiver for the channel matrix $\mathbf{H}$ given as

$$\mathbf{H} = \begin{bmatrix} 2 & 3 \\ 1 & 3 \\ 4 & 2 \end{bmatrix} \quad (6.4)$$

Solution: It can be seen that the MIMO channel matrix above is of size 3×2 , implying that the number of receive antennas is $r = 3$, while the number of transmit antennas is $t = 2$. Thus, the MIMO system model can be described as

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 2 & 3 \\ 1 & 3 \\ 4 & 2 \end{bmatrix}}_{\mathbf{H}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x}} + \underbrace{\begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}}_{\mathbf{n}}$$

Thus, as can be seen from the model above, the transmit vector $\mathbf{x}$ is of dimension 2×1 , while the receive and noise vectors $\mathbf{y}$, $\mathbf{n}$ respectively are of dimension 3×1 . The above MIMO system model can also be explicitly written to describe the signal received at each receive antenna as

$$y_1 = 2x_1 + 3x_2 + n_1$$

$$y_2 = x_1 + 3x_2 + n_2$$

$$y_3 = 4x_1 + 2x_2 + n_3$$

which basically represents a system of $r = 3$ equations for $t = 2$ unknowns x_1, x_2 . To compute the zero-forcing decoder $\mathbf{F}_{ZF}$, we first compute the matrix $(\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$. Observe that $\mathbf{H}^H \mathbf{H}$ can be simplified as

$$\begin{aligned} \mathbf{H}^H \mathbf{H} &= \begin{bmatrix} 2 & 1 & 4 \\ 3 & 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 3 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 21 & 17 \\ 17 & 22 \end{bmatrix} \end{aligned}$$

Hence, the expression for $(\mathbf{H}^H \mathbf{H})^{-1}$ can be simplified as

$$\begin{aligned} (\mathbf{H}^H \mathbf{H})^{-1} &= \frac{1}{21 \times 22 - 17 \times 17} \begin{bmatrix} 22 & -17 \\ -17 & 21 \end{bmatrix} \\ &= \frac{1}{173} \begin{bmatrix} 22 & -17 \\ -17 & 21 \end{bmatrix} \end{aligned}$$

Thus, the zero-forcing receiver matrix $\mathbf{F}_{ZF} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H$ can be expressed as

$$\begin{aligned} \mathbf{F}_{ZF} &= (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \\ &= \frac{1}{173} \begin{bmatrix} 22 & -17 \\ -17 & 21 \end{bmatrix} \begin{bmatrix} 2 & 1 & 4 \\ 3 & 3 & 2 \end{bmatrix} \\ &= \frac{1}{173} \begin{bmatrix} -7 & -29 & 54 \\ 29 & 46 & -26 \end{bmatrix} \\ &= \begin{bmatrix} -0.04 & -0.17 & 0.31 \\ 0.17 & 0.27 & -0.15 \end{bmatrix} \end{aligned} \tag{6.5}$$

Thus, the zero-forcing estimate of the transmit vector is given as

$$\begin{aligned}\hat{\mathbf{x}}_{ZF} &= \mathbf{F}_{ZF} \mathbf{y} \\ &= (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \mathbf{y} \\ &= \begin{bmatrix} -0.04 & -0.17 & 0.31 \\ 0.17 & 0.27 & -0.15 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}\end{aligned}$$

These expressions can now be explicitly written for the estimate of each of the transmit symbols $\hat{x}_1, \hat{x}_2$ as

$$\hat{x}_1 = -0.04y_1 - 0.17y_2 + 0.31y_3$$

$$\hat{x}_2 = +0.17y_1 + 0.27y_2 - 0.15y_3$$

It can also be confirmed that the zero-forcing receiver matrix $\mathbf{F}_{ZF}$ is of dimension 2×3 , i.e., $t \times r$.

One of the chief disadvantages of the zero-forcing receivers is *noise amplification*. This can be understood as follows. Consider the SISO wireless system for $r = t = 1$. We have the system model given as

$$y = hx + n$$

Hence, the zero-forcing receiver is given as $f_{ZF} = (h^* h)^{-1} h^* = h^{-1} = \frac{1}{h}$. Hence, the zero-forcing estimate of the transmitted symbol x is given as

$$\begin{aligned}\hat{x}_{ZF} &= f_{ZF} \times y \\ &= \frac{1}{h} (hx + n) \\ &= x + \frac{n}{h}\end{aligned}\tag{6.6}$$

Observe that if the fading coefficient h is close to zero then the factor $\frac{n}{h}$ is very high. This is termed *noise amplification*, which significantly distorts the performance of the receiver. The MMSE receiver which is proposed next addresses this problem.

3. Explain MIMO MMSE Receiver. (10 marks)

The Minimum Mean-Squared Error (MMSE) receiver is a linear detection scheme for MIMO systems that aims to find an optimal balance between two opposing goals:

- cancelling inter-stream interference** and
- suppressing noise amplification.**

The **Zero-Forcing (ZF)** receiver, which only focuses on **cancelling inter symbol interference**.

The MMSE receiver minimizes the **average total error** between the transmitted and estimated signals, factoring in both the channel characteristics and the noise power.

This makes it more robust and performant than the ZF receiver, **especially at low and moderate Signal-to-Noise Ratios (SNR)**.

The fundamental goal of the MMSE receiver is to find the estimate $\hat{x}_{MMSE}$ that minimizes the **Mean-Squared Error (MSE)**:

$$\mathbb{E} \{ \|\hat{\mathbf{x}}_{MMSE} - \mathbf{x}\|^2 \}$$

Bayesian Approach: The MMSE receiver treats the transmitted signal vector $\mathbf{x}$ as a **random variable** with known statistical properties (e.g., power), unlike the deterministic view sometimes taken for ZF.

Linear Estimator: It restricts itself to a linear estimator of the form $\hat{x} = \mathbf{C}^H \mathbf{y}$ where $\mathbf{C}$ is a receiver matrix to be optimized.

- a) Let us consider SISO system
estimate a *single* transmitted symbol x from a *vector* of received signals
 $y = h x + n$

A linear estimator is defined as:

$$\hat{x} = \mathbf{c}^T \mathbf{y}$$

The goal is to find the vector $\mathbf{c}$ that minimizes $\mathbb{E}\{|\hat{x} - x|^2\}$.

- **Cost Function Expansion:** The Mean-Squared Error (MSE) is expanded into a function of $\mathbf{c}$:

$$\overline{MSE}(\mathbf{c}) = \mathbf{c}^T \mathbf{R}_{yy} \mathbf{c} - 2\mathbf{c}^T \mathbf{R}_{yx} + R_{xx}$$

where:

- $\mathbf{R}_{yy} = \mathbb{E}\{\mathbf{y}\mathbf{y}^H\}$ is the autocorrelation matrix of the received signal.
- $\mathbf{R}_{yx} = \mathbb{E}\{\mathbf{y}x\}$ is the cross-correlation vector between the received signal and the transmitted symbol.
- $R_{xx} = \mathbb{E}\{|x|^2\}$ is the power of the transmitted symbol.
- **Optimization:** By taking the derivative of $\overline{MSE}(\mathbf{c})$ with respect to $\mathbf{c}$ and setting it to zero, the optimal filter is found to be the **Wiener Filter**:

$$\mathbf{c} = \mathbf{R}_{yy}^{-1} \mathbf{R}_{yx}$$

Thus, the MMSE estimate is:

$$\hat{x}_{\text{MMSE}} = \mathbf{R}_{xy} \mathbf{R}_{yy}^{-1} \mathbf{y}$$

- b) Generalize the MIMO: The solution from the SIMO case is then extended to the full MIMO case where a *vector* $\hat{x}$ must be estimated.

The full MIMO system model is:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

The derivation follows the same principle as the SIMO case but with vectors and matrices. Key assumptions are:

- **Transmit Signal Statistics:** Symbols from different antennas are uncorrelated and have equal power P_d .

$$\mathbf{R}_{xx} = \mathbb{E}\{\mathbf{x}\mathbf{x}^H\} = P_d \mathbf{I}_t$$

- **Noise Statistics:** Noise is zero-mean, uncorrelated from the signal, and has variance σ_n^2 .

$$\mathbf{R}_{nn} = \mathbb{E}\{\mathbf{n}\mathbf{n}^H\} = \sigma_n^2 \mathbf{I}_r$$

Using these, the required covariance matrices are computed:

- **Received Signal Covariance:**

$$\mathbf{R}_{yy} = \mathbb{E}\{\mathbf{y}\mathbf{y}^H\} = P_d \mathbf{H}\mathbf{H}^H + \sigma_n^2 \mathbf{I}_r$$

- **Cross-Covariance Matrix:**

$$\mathbf{R}_{yx} = \mathbb{E}\{\mathbf{y}\mathbf{x}^H\} = P_d \mathbf{H}$$

The optimal MIMO MMSE receiver matrix $\mathbf{C}$ is then:

$$\mathbf{C} = \mathbf{R}_{yy}^{-1} \mathbf{R}_{yx} = P_d (\mathbf{H}\mathbf{H}^H + \sigma_n^2 \mathbf{I}_r)^{-1} \mathbf{H}$$

And the final MMSE estimate of the transmitted vector is:

$$\hat{\mathbf{x}}_{\text{MMSE}} = \mathbf{C}^H \mathbf{y} = P_d \mathbf{H}^H (\mathbf{H}\mathbf{H}^H + \sigma_n^2 \mathbf{I}_r)^{-1} \mathbf{y}$$

The Final MMSE Receiver and its Alternative Form

1. Standard Form:

$$\hat{\mathbf{x}}_{\text{MMSE}} = P_d \mathbf{H}^H (P_d \mathbf{H} \mathbf{H}^H + \sigma_n^2 \mathbf{I}_r)^{-1} \mathbf{y}$$

2. Alternative (More Popular) Form:

$$\hat{\mathbf{x}}_{\text{MMSE}} = P_d (P_d \mathbf{H}^H \mathbf{H} + \sigma_n^2 \mathbf{I}_t)^{-1} \mathbf{H}^H \mathbf{y}$$

Why the Alternative Form is Preferred:

- In typical MIMO systems, the number of receive antennas r is greater than or equal to the number of transmit antennas t ($r \geq t$).
- The matrix $\mathbf{H}^H \mathbf{H}$ is $t \times t$, while $\mathbf{H} \mathbf{H}^H$ is $r \times r$.
- Therefore, inverting the $t \times t$ matrix ($P_d \mathbf{H}^H \mathbf{H} + \sigma_n^2 \mathbf{I}_t$) is computationally cheaper than inverting the $r \times r$ matrix, making the alternative form more efficient for implementation.

4. Explain singular value decomposition of MIMO channel. (10 marks)

The Singular Value Decomposition (SVD) is a powerful mathematical tool that decomposes the MIMO channel matrix $\mathbf{H}$ into three simpler, structured matrices: $\mathbf{U}$, Σ , and $\mathbf{V}^H$.

This decomposition reveals that the MIMO channel can be interpreted as a set of parallel, independent, non-interfering sub-channels.

The number of such usable sub-channels is equal to the rank of the channel, and their quality is determined by their singular values.

1. The SVD Equation and Matrix Dimensions

The core decomposition is given by:

$$\mathbf{H} = \mathbf{U} \Sigma \mathbf{V}^H$$

For a channel matrix $\mathbf{H}$ with dimensions $r \times t$ (where $r \geq t$):

- $\mathbf{U}$: An $r \times t$ matrix containing the **left-singular vectors**.
- Σ : A $t \times t$ **diagonal matrix** containing the **singular values** ($\sigma_1, \sigma_2, \dots, \sigma_t$).
- $\mathbf{V}$: A $t \times t$ matrix containing the **right-singular vectors**. $\mathbf{V}^H$ is its Hermitian (conjugate transpose).

2. Key Properties of the Matrices $\mathbf{U}$ and $\mathbf{V}$

The matrices $\mathbf{U}$ and $\mathbf{V}$ have crucial mathematical properties that form the basis for simplifying the MIMO system.

A. Orthonormality of Columns

- The columns of $\mathbf{U}$ (vectors $\mathbf{u}_i$) and $\mathbf{V}$ (vectors $\mathbf{v}_i$) are **unit vectors**:

$$\|\mathbf{u}_i\|^2 = \|\mathbf{v}_i\|^2 = 1$$

- The columns are **mutually orthogonal**:

$$\mathbf{u}_i^H \mathbf{u}_j = \mathbf{v}_i^H \mathbf{v}_j = 0 \quad \text{for } i \neq j$$

Together, these two properties mean the columns are **orthonormal**.

B. Unitarity

- The matrix $\mathbf{V}$ is a **square unitary matrix**. This means:

$$\mathbf{V}^H \mathbf{V} = \mathbf{V} \mathbf{V}^H = \mathbf{I}_t$$

where $\mathbf{I}_t$ is the $t \times t$ identity matrix. A unitary matrix represents a rotation or reflection in complex vector space; it preserves the norm (energy) of any vector it multiplies.

- The matrix $\mathbf{U}$ is **tall** ($r \times t$). It satisfies $\mathbf{U}^H \mathbf{U} = \mathbf{I}_t$, but $\mathbf{U} \mathbf{U}^H \neq \mathbf{I}_r$ unless $r = t$. Such a matrix is called **semi-unitary**.

3. Key properties of the singular values

The diagonal matrix Σ contains the singular values, which are fundamental to understanding MIMO performance.

- Non-negative and Ordered:** The singular values are real and non-negative ($\sigma_i \geq 0$) and are arranged in descending order:

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_t \geq 0$$

- Link to Channel Rank:** The number of **non-zero singular values is equal to the rank of the channel matrix $\mathbf{H}$** .
 - The rank dictates the **maximum number of independent data streams** that can be simultaneously transmitted through the channel.

The SVD provides a profound physical interpretation of MIMO operation:

1. **Precoding (V):** At the transmitter, the data vector $\mathbf{s}$ is multiplied by the matrix $\mathbf{V}$. This "rotates" the transmitted signal to align with the channel's eigenmodes.

2. Parallel Channels (Σ): The channel's effect is now simplified to a set of independent, parallel sub-channels. Each sub-channel has a gain equal to a singular value σ_i . The first sub-channel (gain σ_1) is the strongest, and subsequent ones are weaker.
3. Combining (V^H): At the receiver, the received signal is multiplied by V^H . This "rotates" the signal back, separating the independent data streams perfectly.

This process transforms the complex, coupled MIMO channel into a collection of independent Single-Input Single-Output (SISO) channels, dramatically simplifying the receiver design (no need for complex ZF or MMSE to separate streams from each other).

6. Note on a) SVD and MIMO capacity

c) Alamouti and Space-Time Codes

d) Nonlinear MIMO receiver: V-Blast, MIMO Beamforming.

i) SVD and MIMO capacity

The Singular Value Decomposition (SVD) of the MIMO channel matrix $\mathbf{H}$ is a powerful mathematical tool that reveals the channel's inherent parallel spatial channels or "eigenmodes." By using the SVD for specific linear processing at both the transmitter (**precoding**) and receiver (**combining**), the original coupled MIMO system can be transformed into a set of **independent, parallel Single-Input Single-Output (SISO) channels**.

This process, known as **spatial multiplexing**, allows for the simultaneous transmission of multiple data streams and provides a direct formula for calculating the fundamental **channel capacity** of the MIMO system.

1. The SVD-Based MIMO System Transformation

The standard MIMO system model is:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

The channel matrix $\mathbf{H}$ is decomposed via SVD as:

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

where $\mathbf{U}$ and $\mathbf{V}$ are unitary matrices, and $\mathbf{\Sigma}$ is a diagonal matrix of singular values $\sigma_1, \sigma_2, \dots, \sigma_t$.

The transformation involves two key signal processing steps:

$$\mathbf{y} = \underbrace{\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H}_{\mathbf{H}}\mathbf{x} + \mathbf{n}$$

- a) **Receive Combining:** The receiver processes the received vector $\mathbf{y}$ by multiplying it with the matrix $\mathbf{U}^H$.

$$\tilde{\mathbf{y}} = \mathbf{U}^H \mathbf{y}$$

This aligns the receiver with the channel's spatial directions to separate the data streams.

At the receiver, multiplying by $\mathbf{U}^H$, we have

$$\begin{aligned}\underbrace{\mathbf{U}^H \mathbf{y}}_{\tilde{\mathbf{y}}} &= \mathbf{U}^H (\mathbf{U} \mathbf{\Sigma} \mathbf{V}^H \mathbf{x} + \mathbf{n}) \\ \tilde{\mathbf{y}} &= \underbrace{\mathbf{U}^H \mathbf{U}}_{\mathbf{I}_t} \mathbf{\Sigma} \mathbf{V}^H \mathbf{x} + \underbrace{\mathbf{U}^H \mathbf{n}}_{\tilde{\mathbf{n}}} \\ &= \mathbf{\Sigma} \mathbf{V}^H \mathbf{x} + \tilde{\mathbf{n}}\end{aligned}$$

- b) **Transmit Precoding:** The transmitter pre-processes the data vector $\hat{\mathbf{x}}$ by multiplying it with the matrix $\mathbf{V}$.

This aligns the data streams with the strong spatial directions (eigenmodes) of the channel.

We have $\mathbf{x} = \mathbf{V} \hat{\mathbf{x}}$

$$\begin{aligned}\tilde{\mathbf{y}} &= \mathbf{\Sigma} \mathbf{V}^H \mathbf{x} + \tilde{\mathbf{n}} \\ &= \mathbf{\Sigma} \mathbf{V}^H \mathbf{V} \tilde{\mathbf{x}} + \tilde{\mathbf{n}} \\ &= \mathbf{\Sigma} \tilde{\mathbf{x}} + \tilde{\mathbf{n}}\end{aligned}$$

The above equivalent system model of the MIMO system after the receive and transmit processing operations can be explicitly written as

$$\begin{bmatrix} \tilde{y}_1 \\ \tilde{y}_2 \\ \vdots \\ \tilde{y}_t \end{bmatrix} = \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \vdots \\ \tilde{x}_t \end{bmatrix} + \begin{bmatrix} \tilde{n}_1 \\ \tilde{n}_2 \\ \vdots \\ \tilde{n}_t \end{bmatrix}$$

2. The Equivalent Parallel Channel Model

Substituting the SVD and the processing steps into the system model leads to a simplified equivalent system:

$$\tilde{\mathbf{y}} = \mathbf{\Sigma} \tilde{\mathbf{x}} + \tilde{\mathbf{n}}$$

This can be written explicitly as a set of t independent, parallel equations:

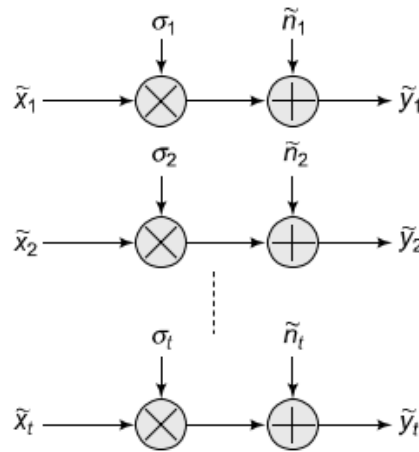
$$\begin{aligned}\tilde{y}_1 &= \sigma_1 \tilde{x}_1 + \tilde{n}_1 \\ \tilde{y}_2 &= \sigma_2 \tilde{x}_2 + \tilde{n}_2 \\ &\vdots \\ \tilde{y}_t &= \sigma_t \tilde{x}_t + \tilde{n}_t\end{aligned}$$

The covariance of the modified $\tilde{\mathbf{n}}$ can be derived as

$$\begin{aligned}\mathbf{R}_{\tilde{\mathbf{n}}} &= \mathbf{E} \{ \tilde{\mathbf{n}} \tilde{\mathbf{n}}^H \} \\ &= \mathbf{E} \{ \mathbf{U}^H \mathbf{n} (\mathbf{U}^H \mathbf{n})^H \} \\ &= \mathbf{E} \{ \mathbf{U}^H \mathbf{n} \mathbf{n}^H \mathbf{U} \} \\ &= \mathbf{U}^H \underbrace{\mathbf{E} \{ \mathbf{n} \mathbf{n}^H \}}_{\sigma_n^2 \mathbf{I}} \mathbf{U} \\ &= \mathbf{U}^H \sigma_n^2 \mathbf{I} \mathbf{U} = \sigma_n^2 \underbrace{\mathbf{U}^H \mathbf{U}}_{\mathbf{I}_t} \\ &= \sigma_n^2 \mathbf{I}_t\end{aligned}$$

The i^{th} MIMO channel is given by

$$\tilde{y}_i = \sigma_i \tilde{x}_i + \tilde{n}_i.$$



3. MIMO Capacity and the "Water-Filling" Principle

The transformation into parallel channels allows for a straightforward calculation of the MIMO channel's **Shannon Capacity**.

- **Capacity of a Single Sub-channel:** The capacity (in bits/sec/Hz) of the i -th parallel channel is:

$$C_i = \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

where P_i is the power allocated to the i -th data stream.

- **Total MIMO Capacity:** The total capacity is the sum of the capacities of all individual parallel channels:

$$C = \sum_{i=1}^t C_i = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

- **Optimal Power Allocation (Water-Filling):** To achieve this maximum capacity, the total transmit power P must be distributed optimally across the sub-channels. The constraint is:

$$P_1 + P_2 + \dots + P_t \leq P$$

The optimal strategy is **Water-Filling**:

1. More power is allocated to stronger sub-channels (those with larger σ_i^2).
2. Less or no power is allocated to weaker sub-channels (those with small σ_i^2 that are in a deep "fade").
3. This algorithm "pours" power into the channels until the total power budget is used, effectively maximizing the sum of the capacities.

ii) Optimal MIMO capacity

For a MIMO system with t parallel subchannels, the **optimal MIMO capacity** problem is formulated as:

$$\max_{P_i} \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

- P_i : Power allocated to the i^{th} subchannel
- σ_i^2 : Gain (or singular value squared) of the i^{th} subchannel
- σ_n^2 : Noise power (per channel)
- P : Total available transmit power
- t : Total number of transmission channels (eigenmodes)

After optimal power allocation, the **maximum achievable capacity** is:

$$C_{\text{opt}} = \sum_{i=1}^t \log_2 \left(1 + \frac{P_i \sigma_i^2}{\sigma_n^2} \right)$$

where P_i are determined by the water-filling solution above.

iii) Asymptotic MIMO capacity

The term “**asymptotic**” here refers to the behavior of the capacity when the number of transmit antennas t becomes **very large** (i.e., $t \gg r$, where r is the number of receive antennas).

- As t increases, the **law of large numbers** implies that $\frac{1}{t} H H^H$ tends to a deterministic matrix.
- This makes capacity analysis simpler and reveals scaling behavior.

$$C_a = (\min \{r, t\}) \log_2 \left(1 + \frac{P_t}{\sigma_n^2} \right)$$

Asymptotic Case

$t \gg r$: capacity simplifies, grows nearly linearly with number of receive antennas

b) MIMO nonlinear Receiver V-blast

V-BLAST is a **nonlinear** MIMO reception strategy

The core idea is to decode the transmitted data streams one by one, and after each successful decoding, to cancel its contribution from the received signal before decoding the next stream. This process reduces interference for the remaining streams.

The standard MIMO system model:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$= \begin{bmatrix} \mathbf{h}_1 & \mathbf{h}_2 & \dots & \mathbf{h}_t \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_t \end{bmatrix} + \mathbf{n}$$

V-BLAST uses the ZF principle as a building block. The ZF receiver matrix $\mathbf{Q}$ is the pseudo-inverse of the channel matrix:

$$\mathbf{Q} = (\mathbf{H}^H \mathbf{H})^{-1} \mathbf{H}^H \quad \text{such that} \quad \mathbf{Q}\mathbf{H} = \mathbf{I}_t$$

Let the rows of $\mathbf{Q}$ be denoted as $\mathbf{q}_1^H, \mathbf{q}_2^H, \dots, \mathbf{q}_t^H$:

$$\mathbf{Q} = \begin{bmatrix} \mathbf{q}_1^H \\ \mathbf{q}_2^H \\ \vdots \\ \mathbf{q}_t^H \end{bmatrix}$$

The property $\mathbf{Q}\mathbf{H} = \mathbf{I}_t$ leads to a crucial orthogonality condition:

$$\mathbf{q}_i^H \mathbf{h}_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

This means that the receive beamformer $\mathbf{q}_i$ is perfectly orthogonal to all channel columns $\mathbf{h}_j$ *except* for $\mathbf{h}_i$. When applied to the received signal $\mathbf{y}$, it will nullify (zero-force) the interference from all streams except the i -th one.

c) MIMO beamforming

(i) **System Model and SVD of the channel**

MIMO Beamforming is antenna technology that uses multiple antennas that create a focused data signal directed at a specific direction, resulting in high SNR. Consider a MIMO system.

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n}$$

$$= \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H \mathbf{x} + \mathbf{n}$$

$$= \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_t \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^H \\ \mathbf{v}_2^H \\ \vdots \\ \mathbf{v}_t^H \end{bmatrix} \mathbf{x} + \mathbf{n}$$

To perform beamforming, we utilize the **Singular Value Decomposition (SVD)** of the channel matrix $\mathbf{H}$. The SVD factorizes $\mathbf{H}$ as:

$$\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^H$$

(ii) **Transmit Beamforming**

The transmitted signal vector is therefore:

$$\mathbf{x} = \mathbf{v}_1 \tilde{x}_1$$

(iii) **Signal Propagation through the Channel**

Now, let's see what happens when this beamformed signal $\mathbf{x}$ passes through the channel $\mathbf{H}$.

$$\begin{aligned} \mathbf{y} &= \mathbf{H}\mathbf{x} + \mathbf{n} \\ &= (\mathbf{U}\mathbf{\Sigma}\mathbf{V}^H)(\mathbf{v}_1 \tilde{x}_1) + \mathbf{n} \end{aligned}$$

Substituting the full SVD matrices:

$$\mathbf{y} = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \dots & \mathbf{u}_t \end{bmatrix} \begin{bmatrix} \sigma_1 & 0 & \dots & 0 \\ 0 & \sigma_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_t \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^H \\ \mathbf{v}_2^H \\ \vdots \\ \mathbf{v}_t^H \end{bmatrix} \mathbf{v}_1 \tilde{x}_1 + \mathbf{n}$$

The critical step is to evaluate $\mathbf{V}^H \mathbf{v}_1$:

$$\begin{bmatrix} \mathbf{v}_1^H \\ \mathbf{v}_2^H \\ \vdots \\ \mathbf{v}_t^H \end{bmatrix} \mathbf{v}_1 = \begin{bmatrix} \mathbf{v}_1^H \mathbf{v}_1 \\ \mathbf{v}_2^H \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_t^H \mathbf{v}_1 \end{bmatrix}$$

Because the columns of $\mathbf{V}$ are orthonormal ($\mathbf{v}_i^H \mathbf{v}_j = 1$ if $i = j$, and 0 otherwise), this simplifies to:

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Substituting this back, the received signal becomes:

$$\begin{aligned} \mathbf{y} &= [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_t] \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_t \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tilde{x}_1 + \mathbf{n} \\ &= [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_t] \begin{bmatrix} \sigma_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \tilde{x}_1 + \mathbf{n} \\ &= \sigma_1 \mathbf{u}_1 \tilde{x}_1 + \mathbf{n} \end{aligned}$$

(iv) Receive Beamforming (Maximal Ratio Combining)

At the receiver, the multiple antennas receive the signal $\mathbf{y}$. To optimally combine these signals, we perform **Maximal Ratio Combining (MRC)**. The optimal combiner is the matched filter to the effective channel, which is $\mathbf{u}_1$.

We compute the decision statistic $\tilde{y}_1$ by projecting the received vector $\mathbf{y}$ onto $\mathbf{u}_1$:

$$\tilde{y}_1 = \mathbf{u}_1^H \mathbf{y}$$

Substituting $\mathbf{y} = \sigma_1 \mathbf{u}_1 \tilde{x}_1 + \mathbf{n}$:

$$\begin{aligned} \tilde{y}_1 &= \mathbf{u}_1^H (\sigma_1 \mathbf{u}_1 \tilde{x}_1 + \mathbf{n}) \\ &= \sigma_1 (\mathbf{u}_1^H \mathbf{u}_1) \tilde{x}_1 + \mathbf{u}_1^H \mathbf{n} \\ &= \sigma_1 \tilde{x}_1 + \tilde{n}_1 \end{aligned}$$

(v) Signal-to-Noise Ratio (SNR) Analysis

The final effective SISO channel is:

$$\tilde{y}_1 = \sigma_1 \tilde{x}_1 + \tilde{n}_1$$

The power of the desired signal is $\sigma_1^2 P$, where $P = E\{|\tilde{x}_1|^2\}$ is the transmit power. The noise power is σ_n^2 .

Therefore, the **SNR at the receiver** is:

$$\text{SNR} = \frac{\sigma_1^2 P}{\sigma_n^2}$$

This shows that the total transmit power P is amplified by the square of the **largest singular value**, σ_1^2 , of the channel matrix. This factor represents the power gain of the dominant spatial mode.

